Lesson Plan

Teacher Name: Ms. Lubna Kakkengal and Mr. Dilan Devendra	**Class/Section:** Grade 4
Subject: TCD: Matatalab Pro Coding	**Date: Week 11 (9 Nov to 13 Nov)**
Class Description: Grade 4 TCD class, Matatalab Pro Session 2 of 4 (1 hour 30 minutes). Students spot repeated steps in a long sequence, group them into a function block and compare the long and short versions of their code.	
TA/LSA Role (if applicable):	

Differentiated Learning Groups (Data)			
Desert Fox	**Camel**	**Oryx**	**Falcon**

Lesson Overview		
Lesson Topic	Function Blocks	
Standard(s)	Technology: decomposition and abstraction (grouping repeated steps); Math: 4.OA.C.5 (generate and analyse patterns)	
Learning Objective(s) (WALT): We are learning to use a function block to make our code shorter and easier to read.		**Success Criteria (WILF):** I can find a pattern of steps that repeats in my code. I can put the repeated steps into a function. I can call the function to run those steps. I can explain why the shorter code is better.
Literacy Strategies	Pattern hunt with coloured cards; sentence frame: My function holds the steps ... and I call it ... times.	
Assessment	Before and after code comparison on the worksheet; observation; live quiz score on the TCD website.	
Targeted Vocabulary	function, pattern, repeat, call, sequence, efficient, debug	
Skills	Pattern recognition, abstraction, logical thinking, teamwork	
National Identity Integration	Repeating patterns in Sadu weaving: Emirati weavers repeat the same small pattern to build a long, beautiful design, just like a	

	function repeats steps.
AI/Digital Learning	Matatalab Pro Sets; hook video; live quiz on the TCD website. Presentation: Grade4_Week11_Presentation.pptx. Worksheet: Grade4_Week11_Worksheet.

Lesson Structure	
Starter (Hook) Engage and Ask (10 minutes)	Warm-up: Clap the Pattern (slide 4). Show the hook video and ask the hook question (slide 2). Share the learning goals (slide 3) and key words (slide 5).
I Do (20 minutes)	Explain what a function is and how to call it (slide 6). Model finding the pattern in a long code using coloured cards (slide 7). Build the square with a function and compare block counts (slides 8 and 9).
We Do (15 minutes)	Class looks at a long staircase code on the board and agrees which steps repeat. Pairs build the staircase function together with teacher support.
You Do (35 minutes)	Pairs complete the function challenges and record long and short code (slide 11). Challenge levels on the board (slide 12). Live quiz on tablets (slide 13).
Differentiated Instruction: **Beginning / On / Above**	**Beginning: Use the function card to draw the square, copying the example.** **On: Create functions for the square and the staircase yourself.** **Above: Use a function together with a loop and draw a pattern of two squares.**
Closure/ Reflection (10 minutes)	Pairs share their block count before and after. Exit ticket: why is a function useful? (slide 14)

Inclusive Planning
Students receiving additional inclusive support (IEP or ILPs) or Gifted and Talented: EAL: colour-coded pattern cards and sentence frame. SEN: worked example card for the square; teacher checks after each step. G&T: function inside a loop to draw two squares.
Inclusion / EAL Support

☑ Visual Supports
☑ Chunked Instructions
☐ Reduced Language Load
☑ Model / Worked Example
☐ Pre-teach Vocabulary
☐ TA/LSA Support
☐ Extra Processing Time
☐ Text Read by TA/LSA
☐ Flexible Grouping
☐ Assistive Technology
☐ Movement Break
☑ Scaffold / Sentence Starters
☐ Other:

G&T / Extension

☑ Extension / Challenge Task
☑ Higher-order Questioning
☐ Depth & Complexity Task
☐ Independent Inquiry
☐ Leadership / Peer Coaching
☐ Real-World Application
☐ Curriculum Compacting
☐ Above-grade-level Texts
☐ Student-led / Passion Project
☐ Open-Ended Task
☐ Other:

Assessment Adaptations

☑ Oral Response
☐ Reduced Questions
☑ Scaffolded Success Criteria
☐ Extra Time
☐ Alternative Outcome Format
☑ Teacher Conferencing/Live Feedback
☐ Quiet / Separate Space
☐ Other: